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QUICK REFERENCE

ODT-REF

USAREUR-AF

LL-001: AVT25 Assessment Tools -- Exponential Work Multiplication

Quick Reference

HEADQUARTERS
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26 MARCH 2026

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sidebar_position: 2 title: "Lessons Learned" description: How five tools in the same enclave multiplied work exponentially by failing to share doctrinal primitives layer: architecture category: reference

LL-001: AVT25 ASSESSMENT TOOLS -- EXPONENTIAL WORK MULTIPLICATION

ODT Lessons Learned Series -- LL-001

Five tools in the same enclave, all implementing the same doctrinal concepts independently. Zero shared primitives. Work multiplying exponentially with every tool added.

Field	Value
Series	LL-001
Exercise	AVT25
Severity	Critical
Category	Architecture -- Shared Primitives
Doctrine Ref	ADP 5-0 (The Operations Process)

BOTTOM LINE UP FRONT

Five assessment tools in the **same enclave** each independently implemented the same three doctrinal primitives (Assessment, Finding, Recommendation), resulting in **15 separate implementations** of the same concepts. Analysts spend **3 hours per assessment** re-entering the same data across tools. Bug fixes take **30 hours** instead of 6. Cross-tool queries are impossible without manual Excel aggregation. **This is not a cross-domain problem. This is a failure to build shared primitives from doctrine.**

01 -- THE PROBLEM IN NUMBERS

5 tools in the SAME ENCLAVE, all performing assessment per ADP 5-0:

1. Strategic Assessments App

2. IMO/LOO Tool
3. Product Tracker
4. Problem Intake Form
5. Lessons Learned System

Each tool independently implemented the same three doctrinal primitives:

- Assessment
- Finding
- Recommendation

5 tools x 3 primitives = 15 separate implementations

02 -- TIME COST PER ASSESSMENT

Before DBO -- AVT25 Pattern

Step	Task	Time
1	Conduct assessment	60 min
2	Enter in Strategic Assessments App	30 min
3	Re-enter in IMO/LOO Tool	30 min
4	Re-enter in Product Tracker	20 min
5	Re-enter in Problem Intake Form	20 min
6	Re-enter in Lessons Learned	20 min
Total		**180 min (3 hrs)**

Data quality degrades with each re-entry: typos introduced, different phrasing causes confusion, fields omitted due to fatigue, inconsistent severity assessments.

After DBO -- Target

Step	Task	Time
1	Conduct assessment	60 min
2	Enter once using DBO primitives	5 min
Total		**65 min**

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64% reduction -- 115 min saved per assessment. Single entry, zero transcription errors. Consistent across all tools. All fields captured once.

03 -- THE SCALING PROBLEM

What happens when we add a 6th tool?

Before DBO -- Exponential O(n^2):

- New tool implements Assessment, Finding, Recommendation
- Now 6 x 3 = **18 implementations**
- Analyst workflow: 6 tools x 30 min = **210 min (3.5 hrs)**
- Maintenance burden: fix bugs 6 times
- Each new tool adds burden to ALL existing tools. Each new primitive must be replicated n times. Cross-tool queries become UNION across n tables.

After DBO -- Linear O(n):

- New tool consumes existing Assessment primitives
- Still **3 implementations** (shared backend)
- Analyst workflow: still **65 min**
- Maintenance burden: fix once
- New tools are additive, not multiplicative. Primitives remain constant. Cross-tool queries are one SELECT.

Scaling Projections

Metric	Before (n=5)	Before (n=10)	After DBO	Savings
Primitive implementations	15	30	3	80-90%
Time per assessment	180 min	330 min	65 min	64-80%
Time per bug fix	30 hrs	60 hrs	6 hrs	80-90%
Time per feature addition	52 hrs	102 hrs	11 hrs	79-89%
Cross-tool query	2-4 hrs manual	4-8 hrs manual	30 sec	>99%

Pattern: Savings increase as n increases. The DBO approach gets *better* with scale while the AVT25 pattern gets exponentially worse.

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04 -- QUERY CAPABILITY

Task: "Show me all critical findings from the last 30 days."

Before DBO (AVT25)

```
-- IMPOSSIBLE without manual aggregation
-- 1. Query Strategic Assessments App database
-- 2. Export to Excel
-- 3. Query IMO/LOO Tool database
-- 4. Export to Excel
-- 5. Query Product Tracker database
-- 6. Export to Excel
-- 7. Query Problem Intake database
-- 8. Export to Excel
-- 9. Query Lessons Learned database
-- 10. Export to Excel
-- 11. Manually combine 5 Excel files
-- 12. Deduplicate (same finding entered 5 times)
-- 13. Format for leadership brief
```

Time: 2-4 hours of manual work. Result: Inconsistent, error-prone.

After DBO

```
SELECT
    a.title,
    af.finding_text,
    af.severity,
    tac.tool_id AS source_tool,
    a.assessed_by,
    a.assessment_date
FROM assessments a
JOIN assessment_findings af
    ON a.assessment_id = af.assessment_id
LEFT JOIN tool_assessment_contexts tac
    ON a.assessment_id = tac.assessment_id
WHERE af.severity = 'critical'
    AND a.assessment_date >= CURRENT_DATE - INTERVAL '30 days'
ORDER BY a.assessment_date DESC;
```

Time: 30 seconds. Result: Comprehensive, accurate, real-time.

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05 -- MAINTENANCE COST

Bug Fix: "Assessment date validation is wrong"

Before DBO:

Step	Task	Time
1	Fix in Strategic Assessments App	6 hrs
2	Fix in IMO/LOO Tool	6 hrs
3	Fix in Product Tracker	6 hrs
4	Fix in Problem Intake	6 hrs
5	Fix in Lessons Learned	6 hrs
Total		**30 hours**

After DBO:

Step	Task	Time
1	Fix in Assessment primitive	6 hrs
	All 5 tools inherit the fix automatically	
Total		**6 hours**

80% reduction -- 24 hours saved.

Feature Addition: "Add confidence level to assessments"

Before DBO:

Step	Task	Time
1	Update 5 schemas + UIs	40 hrs
2	Migrate data in 5 databases	12 hrs
Total		**52 hours**

After DBO:

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Step	Task	Time
1	Add to Assessment primitive + backend	4 hrs
2	Migrate data once	2 hrs
3	Update 5 tool UIs to display field	5 hrs
Total		**11 hours**

79% reduction -- 41 hours saved.

06 -- THE MATH OF EXPONENTIAL WORK

```
n = number of tools
p = number of primitives (Assessment, Finding, Recommendation)
t = time per entry
m = maintenance time per change

Before DBO (AVT25 Pattern):
  Implementations  = n x p      (grows with both dimensions)
  Entry time       = n x t      (analyst enters n times)
  Maintenance      = n x m      (fix in n places)
  Queries          = O(n)       (UNION across n tables)

After DBO:
  Implementations  = p          (constant, regardless of n)
  Entry time       = t          (analyst enters once)
  Maintenance      = m          (fix once)
  Queries          = O(1)       (single SELECT)
```

07 -- LEADERSHIP STATEMENTS

Propagating these tools without a shared backend does not scale linearly — it multiplies the work exponentially. Every new assessment tool added under the AVT25 pattern increases burden across all existing tools. This is mathematically unsustainable.

Lesson: Every new assessment tool adds burden to ALL existing tools. The work does not scale linearly — it scales exponentially.

The distinction between what end users request and what doctrine requires must be maintained. Doctrine is the authoritative starting point. Do not derive Assessment object definitions from individual tool team preferences.

Lesson: Do not ask each tool team "what Assessment object do you want?" (results in 5 different answers). Ask ADP 5-0 "what is Assessment?" (one answer). Build it once. All tools consume it.

08 -- THE SOLUTION -- DBO CONVERGENCE

Phase 1 -- 30 Days: Build Assessment Primitives from ADP 5-0 (Foundation)

- Assessment (identity)
- AssessmentFinding (identity)
- AssessmentRecommendation (identity)
- ToolAssessmentContext (classification)

Phase 2 -- 90 Days: Migrate Strategic App + IMO/LOO Tool (Proof of concept)

- Demonstrates shared backend working
- Proves 64% time savings
- Validates cross-tool queries

Phase 3 -- 90 Days: Migrate Remaining 3 Tools (Full convergence)

- Product Tracker
- Problem Intake Form
- Lessons Learned System

Result

Dimension	Before	After
Backends	5 tools -> 5 backends	5 tools -> 1 backend
Analyst time	180 min / assessment	65 min / assessment
Scaling behavior	Exponential	Linear
Cross-tool queries	Manual Excel aggregation	Single SQL query
Sustainability	Unsustainable	Sustainable

BOTTOM LINE

AVT25 Assessment Tools are the smoking gun.

Same enclave. Same doctrinal function. Zero shared foundation. Work multiplying exponentially.

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This is not a cross-enclave problem (CDS). This is not a cross-workshop problem (doctrine drift). **This is a fundamental failure to build shared primitives from doctrine.**

If we cannot get assessment tools in the same enclave to share a backend, we have already lost.

Doctrine is gospel. Start there.

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LL-002: DATA LITERACY BASELINE REQUIRED BEFORE ODT EMPLOYMENT

ODT Lessons Learned Series -- LL-002

Deploying specialized ODTs to units without a data literacy baseline wastes ODT capacity and produces minimal operational value. Units must possess foundational skills before ODT integration.

Field	Value
Series	LL-002
Exercise	Mojave Falcon (Army Reserve, Jun 2025)
Severity	Significant
Category	Training -- Prerequisite Enforcement
Doctrine Ref	MCCoE Decision Optimization CONOPS; ADP 7-0 (Training)

BOTTOM LINE UP FRONT

The Army Reserve's first ODT deployment at Exercise Mojave Falcon (Fort Hunter Liggett, June 2025) demonstrated that **ODTs are not a substitute for baseline training**. Staff who had not trained on data tools before the exercise refused to learn them during operations. Leaders who lacked data literacy could not articulate analytical requirements, resulting in ODT capacity wasted on ad-hoc requests rather than structured problem-solving.

01 -- THE PROBLEM

The OCAR Chief Data and Analytics Office deployed an ODT supporting the 79th TSC and 311th ESC. The team brought Palantir, Tableau, and Power BI capabilities. Key friction points:

- **Cultural resistance:** Some staff preferred "map boards and acetate overlays." One intelligence officer stated they were "just calling all routes red because we don't really see a pattern" — demonstrating a fundamental analytical gap no ODT can fix on the fly.
- **Training gaps:** Personnel had not trained on tools beforehand. One officer said: "This is great, but I don't have time to figure out a new tool while we're trying to execute."
- **Data governance confusion:** Personnel unclear on responsibilities for managing digital artifacts. Export from classified to unclassified required approvals nobody had pre-coordinated.

- **Simulation data quality:** Exercise data poorly mimicked real-world conditions — ammunition lacked TAMIS-level granularity; food and fuel information was oversimplified.

02 -- WHAT WORKED

Early adopters — future battalion and brigade commanders, CW5s, and SGMs — embraced the ODT, requested data analysis, and leaned into predictive capabilities. These leaders represent the formation's data-literate nucleus. The ODT found that elevating the quality of questions staff asked produced more value than delivering finished products.

03 -- LESSON

ODTs should deploy to units that already possess a data literacy baseline. Deploying a specialized analytics team to a unit with no digital muscle memory is not a great use of resources. This is an institutional culture challenge, not a technology limitation.

Application to MSS Training: - TM-10 (Maven User) and TM-20 (Builder) establish the minimum baseline. Units must complete TM-10 theater-wide before ODT integration produces full value. - The Unit Data Trainer (T3-F) model builds local capacity between MTT visits — exactly the mechanism needed to create the baseline that ODTs require. - TM-30 is the hard prerequisite before specialist employment. Mojave Falcon validates this: personnel without TM-30-level skills cannot effectively consume or direct ODT products.

Source: Perkins, Jim. "Lessons from the Army Reserve's First Operational Data Team." War on the Rocks, July 2025.

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LL-003: COMPLEMENTARY ARMY DATA LITERACY PROGRAMS

ODT Lessons Learned Series -- LL-003

Multiple Army organizations are simultaneously developing data literacy training. Awareness of complementary programs prevents duplication and enables cross-referencing.

Field	Value
Series	LL-003
Severity	Informational
Category	Training -- Institutional Awareness
Doctrine Ref	ADP 7-0 (Training); TR 350-70

PROGRAMS IDENTIFIED

Program	Proponent	Format	Audience	Relevance to MSS
Signal School Data for Leaders Course	U.S. Army Signal School (Fort Eisenhower)	4-day MTT, Power BI focused	Senior military leaders	Establishes commander data literacy baseline. Complementary to TM-EXEC. Created by CPT Derek Koslowski, CDO at Signal School. First MTT delivered at Fort Cavazos, Nov 2024. Topics: data literacy, data governance, cloud fundamentals, storytelling with data, zero trust.
AKMP Data Immersion Course	MCCoE/C2ID (AKMP)	32-hour online	KMs at DIV+ level	Covers data governance, Army Data Catalog, data workforce roles, GenAI, Vantage. Aligns with TM-40K.
Data Literacy 101 (DL101-TTC)	West Point CDAS	5-day TTC, 10 hrs instruction	Train-the-Trainer	Foundational data literacy + teaching methodology. 140 attendees at Jun 2024 iteration.
TRADOC OCKO Data Literacy Portal	TRADOC C2DAO	Online courses	TRADOC workforce	Central hub for data literacy training. Decision Optimization Branch targeting FOC Dec 2025.

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Program	Proponent	Format	Audience	Relevance to MSS
CALL 25-10	Center for Army Lessons Learned	Handbook (PDF)	Commanders and staff	Data literacy fundamentals tailored for military leaders. Published Apr 2025.
CAC Maven Integration	MCCoE/CGSC/CAC C2DAO	8-hr operator course + PME integration	CGSC students, operators	Maven being integrated into KM Qual Course, SAMS, SCP, and Data Academy Builders Course. Train-the-trainer held Feb 2026.

BOTTOM LINE

The MSS Training Program is ahead of the institutional curve — our curriculum is published, delivering, and evaluated to Go/No-Go standard while many of these programs are still in pilot. Awareness of these programs enables cross-referencing in supplementary reading sections and validates the MSS training design against institutional Army direction.

Distribution: Authorized personnel -- USAREUR-AF and supporting commands Proponent: Operational Data Team, USAREUR-AF Authority: Chief Data and Analytics Officer, USAREUR-AF Classification Guide: Per DoDI 5200.48 and DoDM 5200.01, Vol 2 CUI

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